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Low reaction force drill systems for lunar anchoring

Low Reaction Force Drill Systems for Lunar Surface Anchoring: Methods, Physics, Materials, and Integration with Inflatable Composite Masts

Quick Reference

Key Findings Table

Theme	Summary	Supporting Citations
Diverse Drilling Methods	Percussive, rotary-percussive, auger-based, and biomimetic anchoring methods minimize reaction forces; each has unique trade-offs in efficiency and regolith removal.	1 2 3
Lunar Environment Physics	Low gravity reduces downforce needs, but increased regolith cohesion and poor thermal dissipation require careful optimization of tool geometry and vibration parameters.	4 5 6
Material and Coating Innovations	Polycrystalline diamond compact bits, advanced coatings (DLC, WS2), and regolith-resin composites improve durability and performance under lunar conditions.	7 8 9
Feedback and Control	Sensor feedback and active damping are essential for managing flexible mast deflections and drill stability.	10 11
Integration Challenges	EMI management and mechanical coupling are critical for reliable operation of co-located drill and communication hardware.	12 13

Direct Answer

The low reaction force drill system for lunar anchoring employs methods designed to minimize the mechanical loads transferred to the lander, an imperative demand in a low-gravity environment. Techniques include percussive drills with either attached or rebounding hammers, rotary-percussive drills that combine rotation with impact energy delivery, and auger-based systems which use helical grooves for efficient regolith removal. The physics of these methods is heavily influenced by the Moon’s reduced gravity, which lowers the baseline reaction forces but is counterbalanced by increased regolith cohesion (partly due to van der Waals forces) and unique soil-tool interaction dynamics. Optimizing cutting geometry (such as rake and side rake angles) and controlling parameters like percussive frequency and hammer dynamic stiffness are key to enhancing impact energy transfer while minimizing reaction forces. Materials such as polycrystalline diamond compact (PDC) bits, coupled with advanced composite coatings (for example, DLC or WS2 impregnated microstructures), are promising for managing wear, thermal stresses, and abrasive lunar dust. In addition, integrating real-time sensor feedback and advanced control systems can ensure stability amid the cross-coupled dynamics of drilling and structural flexing when mounted on inflatable composite masts.

Study Scope

- **Time Period:** Recent advances (last 10–15 years)
- **Disciplines:** Mechanical engineering, materials science, lunar geotechnics, robotics, electromagnetic compatibility
- **Methods:** Meta-analysis, simulation, experimental studies, discrete element modeling, additive manufacturing, control systems engineering

Assumptions & Limitations

- Most findings are based on lunar regolith simulants and analog environments; in-situ lunar validation is limited.
- Integration studies for EMI and mechanical coupling are largely conceptual; experimental data is sparse.
- Material performance under extreme lunar thermal cycling and vacuum is extrapolated from laboratory tests.

Suggested Further Research

- Experimental validation of optimized percussive frequencies and novel drill bit designs in lunar regolith simulants.
- Integration studies focusing on EMI management and mechanical coupling between inflatable structures and low reaction force drilling systems.
- Development of adaptive, self-tuning control algorithms for real-time drill and mast stabilization.
- In-situ additive manufacturing and repair of drill components using lunar regolith.

1. Introduction

Background and Motivation

Lunar surface anchoring is a critical challenge for lightweight landers and infrastructure, especially as missions transition from exploration to sustained operations. The Moon’s regolith presents unique mechanical and environmental challenges: low gravity reduces the downforce available for anchoring, while increased soil cohesion and abrasive dust complicate drilling and structural stability. Inflatable composite carbon fiber masts offer scalable, lightweight solutions for deploying communication hardware and solar panels, but require rigid anchoring to function reliably. Low reaction force drill systems are thus essential—they enable secure anchoring without imposing excessive loads on the lander, facilitating integration with inflatable masts and sensitive electronics [1] [4] [14].

2. Low Reaction Force Drilling Techniques for Lunar Anchoring

Overview of Drilling Methods

- **Percussive Drilling:** Utilizes impact energy to fragment regolith, significantly reducing required downforce (up to 15×) compared to conventional methods. Two main mechanisms:
  - **Attached Hammer:** Hammer is fixed to the drill stem, producing higher stress magnitudes and frequencies.
  - **Rebounding Hammer (Down-the-Hole):** Hammer rebounds, optimizing mass matching and energy transfer, especially effective in shallow regolith [2] [15] [16].
- **Rotary-Percussive Drilling:** Combines rotation and percussion, improving penetration in hard materials and facilitating cuttings removal. Requires careful management of power and reaction forces [3] [17].
- **Auger-Based Drilling:** Helical grooves transport regolith efficiently, supporting low reaction force operation and sampling [3].
- **Biomimetic Anchoring:** Inspired by plant root growth, these compliant, low-energy anchors distribute friction and control, potentially offering robust anchoring with minimal reaction forces [16].
- **Helical Screw-In Anchors:** Provide lateral resistance, with performance dependent on depth, shaft size, and plate diameter [18].

Synthesis

Each method offers distinct advantages and trade-offs. Percussive and rotary-percussive systems are most mature for lunar anchoring, while auger and biomimetic approaches offer promising alternatives for specific regolith conditions and mass constraints [1] [2] [3].

Percussive and Rotary-Percussive Drilling

- **Efficiency:** Down-the-hole and hammer drilling achieve >90% efficiency; attached hammer mechanisms produce higher stress but may require more energy. Mass matching and impact frequency optimization are critical [19] [20].
- **Reaction Force Reduction:** Down-the-hole hammers transmit energy more efficiently, reducing reaction forces, especially in shallow drilling. Rotary-percussive drills further reduce required weight on bit, crucial under lunar gravity [21] [22].
- **Design Trade-offs:** Rotary-percussive drills consume more power but offer faster penetration and better cuttings removal. Pure percussive drills are lighter and simpler but may struggle with cuttings transport [22] [23].

Synthesis

Optimizing hammer mass, impact frequency, and tool geometry is essential for maximizing efficiency and minimizing reaction forces. Rotary-percussive systems are favored for deeper or harder regolith, while percussive-only systems excel in shallow, unconsolidated soils [19] [20] [22].

Optimization of Cutting Parameters

- **Rake Angle:** Increasing rake angle reduces penetration but increases resistive forces; optimal angles balance fragmentation and force reduction [24] [25].
- **Percussive Frequency:** Higher frequencies reduce drilling load and improve penetration, but optimal ranges depend on regolith properties and tool design [26] [27].
- **Tool Geometry:** Attack angle and side rake influence resistive forces; discrete element modeling enables parameter optimization [15] [28].

Synthesis

Simultaneous optimization of rake angle, percussive frequency, and tool geometry is necessary for efficient, low-force drilling in lunar regolith. Real-time monitoring and adaptive control can further enhance performance [15] [24].

Novel Drill Bit Geometries and Materials

- **Bit Design:** Conical bits with multi-row serrated tines and ultrasonic/sonic driller/corer mechanisms optimize cutting mechanics and minimize reaction forces [29] [30].
- **Materials:** Polycrystalline diamond compact (PDC) bits, advanced coatings (DLC, WS2), and nanostructured composites improve durability and energy transfer [7] [8].
- **Adaptive Control:** Self-tuning algorithms and real-time force prediction enable dynamic optimization of drilling parameters [31] [32].

Synthesis

Innovative bit geometries and advanced materials are pivotal for reducing reaction forces and enhancing drill longevity under lunar conditions. Adaptive control strategies further optimize energy transfer and system stability [7] [30].

3. Physics of Low Reaction Force Drilling in Lunar Gravity

Soil-Tool Interaction in Low Gravity

- **Gravity Effects:** Reduced lunar gravity lowers reaction forces and energy consumption, enabling lighter systems. However, increased regolith cohesion and van der Waals forces counteract these benefits, making soil cutting more challenging [4] [5].
- **Penetration Mechanics:** Depth dependence of resistance is reduced; granular flow and rate-dependent behavior necessitate new rheological models [33].
- **Optimization:** Rotation speed, vibration, and tool geometry can further reduce excavation forces by up to 50% [34] [35].

Synthesis

Low gravity fundamentally alters soil-tool interaction, requiring tailored drilling strategies and parameter optimization to achieve low reaction forces and efficient penetration [4] [5].

Role of Van der Waals Forces and Soil Cohesion

- **Cohesion:** Van der Waals forces significantly increase regolith cohesion and frictional resistance, especially under low confining pressures [36] [37].
- **Cutting Resistance:** Energy consumption and bending moments rise due to enhanced soil strength; neglecting these forces leads to unsafe excavation estimates [5] [38].
- **Quantitative Impact:** Internal friction angles and apparent cohesion are higher in lunar regolith, affecting drilling resistance and tool design [39] [40].

Synthesis

Van der Waals forces are a dominant factor in lunar regolith mechanics, necessitating their inclusion in soil-tool interaction models and drill design for accurate force prediction and safe operation [36] [37].

Thermal and Mechanical Stresses in Lunar Vacuum

- **Heat Dissipation:** Ultra-high vacuum impedes convective cooling, causing significant temperature rise and thermal stress on drill bits [6] [41].
- **Material Response:** PDC bits show superior stability and lower torque under vacuum; cemented carbide bits require optimization [5] [7].
- **Vibration Control:** Absence of air damping leads to sustained oscillations; advanced damping methods are needed [42].

Synthesis

Thermal and mechanical stresses in lunar vacuum demand robust materials and strategic drilling modes (e.g., drilling-pause-drilling) to prevent tool failure and maintain operational reliability [6] [7].

Vibration-Assisted Drilling and Optimal Parameters

- **Ultrasonic Vibration:** Resonance at ~20 kHz and amplitudes of 10–40 μm effectively reduces resistive forces, independent of gravity [43].
- **Force Reduction:** Vibratory tools decrease excavation force by up to 10×, enabling smaller, more efficient machinery [44] [45].
- **Parameter Optimization:** Optimal frequencies and amplitudes must be identified for each regolith type; excessive vibration can increase damage [46] [47].

Synthesis

Vibration-assisted drilling is a powerful tool for minimizing reaction forces in lunar regolith, but requires careful parameter tuning to avoid excessive wear or damage [43] [46].

4. Materials and Design Considerations for Lunar Drill Components

Material Selection for Drill Bits and Structural Elements

- **PDC Bits:** Offer superior stability, low torque, and high penetration speed; optimal for lunar drilling [7].
- **Geopolymer Concretes:** High compressive strength and durability, but brittle; fiber reinforcement improves ductility [48] [49].
- **Regolith-Resin-Fiber Composites:** Retain high compressive strength under lunar conditions; better flexural and tensile strength retention than resin-only composites [50].
- **Aluminum Oxide Coatings:** Provide wear resistance and dust mitigation [51].

Synthesis

Material selection must balance strength, toughness, wear resistance, and thermal stability. PDC bits and fiber-reinforced geopolymers are leading candidates for drill and structural elements [7] [49].

Coatings and Surface Treatments for Dust and Thermal Protection

- **PDC with MoS_2 Impregnation:** Reduces friction and wear under vacuum and thermal cycling [8].
- **DLC Coatings:** Enhance hardness and reduce friction; microtexturing improves adhesion [52] [53].
- **WS_2 Coatings:** Synergistic with laser surface texturing for improved wear resistance [54].
- **Polymer Composites:** PEEK and ATSP coatings mitigate abrasive effects and reduce friction [55].

Synthesis

Advanced coatings and surface treatments are essential for drill durability and dust mitigation. Bionic self-lubricating PDC and DLC-coated microtextured surfaces are particularly promising [8] [52].

Additive Manufacturing and In-Situ Repair

- **Material Extrusion (MEX):** Recycled PEKK with regolith enables closed-loop fabrication and repair [9].
- **Laser Powder Bed Fusion (LPBF):** Facilitates in-situ manufacturing of complex parts; process optimization needed for vacuum and strength [56].
- **ISRU:** Reduces payload mass and enables sustainable repair and manufacturing [57] [58].

Synthesis

Additive manufacturing and ISRU are transformative for lunar missions, enabling on-demand repair and fabrication of drill components, reducing reliance on Earth-supplied materials [9] [57].

Ultrasonic Vibration-Assisted Machining of Composites

- **UAD:** Reduces cutting temperature by up to 50% and axial force by 40%, producing delamination-free holes in fiber composites [59].
- **Horizontal Ultrasonic Vibration:** Minimizes delamination and improves hole quality in CFRP [60].
- **Robotic UAD:** Adapts to tool wear, reducing delamination and cutting-edge rounding [61].
- **Cryogenic RUM:** Further reduces thrust force and surface roughness [62].

Synthesis

Ultrasonic vibration-assisted machining is highly effective for minimizing thermal damage and wear in composite carbon fiber masts, supporting high-quality integration with lunar landers [59] [61].

5. Integration with Inflatable Composite Masts and Communication Hardware

System Integration Challenges and Solutions

- **Borehole Stability:** Unknown regolith and bedrock require lightweight casing solutions; ISRU-based casing generation reduces mass and enhances stability [63].
- **Inflatable Masts:** Require advanced stabilization and feedback control to manage tip deflections and payload orientation [14] [64].
- **Electromagnetic Compatibility:** EMI management is critical; advanced materials and computational design tools are needed [65].
- **Software and Procedural Harmonization:** Essential for safe, reliable operations among international partners [66] [67].

Synthesis

Integration demands multidisciplinary solutions—mechanical, electromagnetic, and procedural. Modular, scalable designs and advanced control systems are key to successful deployment [14] [63].

Lightweight Casing and Borehole Stabilization

- **Vacuum-Solidifying Reinforcement Fluids:** Increase cohesion and improve borehole stability; optimal solute concentration is 0.15–0.25 g/mL [68].
- **Ultra-Lightweight Cement Slurry:** Incorporates aluminosilicate microspheres for low-density, homogeneous casing [69].
- **Polymer-Based Reinforcement Liquids:** Solidify in vacuum and low temperature, pre-placed in drill pipes [68].

- **ISRU-Based Casing Generation:** Reduces payload mass and power requirements, enabling sustainable casing solutions [63].

Synthesis

Lightweight, ISRU-enabled casing materials and vacuum-solidifying fluids are essential for maintaining borehole stability during low reaction force drilling on the Moon [63] [68].

Electromagnetic Interference Mitigation

- **Conductive Coatings and Shielding:** Prevent EMI/RFI, especially with plastic enclosures [12].
- **Grounding and Packaging:** Comprehensive EMI control plans are critical [70].
- **Spatial and Frequency Diversity:** Mitigate multipath fading and interference [71].
- **Capacitive Segmentation and Flexible PCB Shields:** Block capacitive coupling while preserving inductive signal detection [13].
- **Polymer Composites with Magnetic Fillers:** Lightweight, efficient EMI shielding for space applications [72].

Synthesis

EMI mitigation requires integrated design strategies—shielding, grounding, and advanced materials—to ensure reliable operation of co-located drill and communication hardware [12] [13].

Stabilization and Feedback Control of Inflatable Masts

- **Hybrid Feedback Control:** Open- and closed-loop strategies manage nonlinearities and external disturbances [73].
- **LQR-Type Controllers:** Limit tip deflection and reduce actuator forces [74].
- **Multiloop Balanced Bridge Feedback:** Improves pointing accuracy and decouples motor loops from flexible plant dynamics [10].
- **Strain Gauge Feedback:** Offers robustness against payload variations [75].
- **Active Damping Controllers:** Achieve targeted damping for multiple structural modes [11].

Synthesis

Advanced feedback and stabilization methods are vital for maintaining mast orientation and structural integrity, especially under lunar environmental stresses [10] [73].

ISRU Integration and Mass Optimization

- **Modular ISRU Architectures:** Enable scalable, long-term stability and adaptability [76] [77].
- **3D Printing with Regolith:** Minimizes mass delivered from Earth, ensures structural integrity [78].
- **Robotic and Automated Processes:** Reduce human intervention, enable continuous drill anchoring and casing generation [79].
- **System Design:** Must address abrasive regolith, extreme temperatures, dust, static electricity, and low gravity [80].

Synthesis

ISRU-based casing and anchoring systems, integrated with compact landers, are essential for mass optimization and long-term stability of lunar infrastructure [78] [81].

6. Conclusion and Future Directions

Summary of Key Insights

- Low reaction force drilling systems leverage percussive, rotary-percussive, and auger-based methods to minimize mechanical loads and enable lightweight lunar anchoring.
- The unique lunar environment—low gravity, high regolith cohesion, ultra-high vacuum—necessitates optimization of tool geometry, vibration parameters, and material selection.
- Polycrystalline diamond compact bits, advanced coatings, and regolith-resin composites are leading material solutions for durability and performance.
- Integration with inflatable composite masts and communication hardware requires advanced feedback control, EMI mitigation, and lightweight casing strategies.
- ISRU and additive manufacturing are transformative for sustainable lunar operations, enabling in-situ repair and fabrication.

Recommendations for Future Research

- Experimental validation of optimized percussive frequencies and novel drill bit designs in lunar regolith simulants.
- Integration studies focusing on EMI management and mechanical coupling between inflatable structures and low reaction force drilling systems.
- Development of adaptive, self-tuning control algorithms for real-time drill and mast stabilization.
- In-situ additive manufacturing and repair of drill components using lunar regolith.
- Exploration of biomimetic anchoring and ISRU-enabled casing generation for scalable lunar infrastructure.

Synthesis

Low reaction force drilling systems for lunar anchoring represent a convergence of advanced mechanical, material, and control engineering, tailored to the Moon's unique environment. Percussive and rotary-percussive methods, optimized through adaptive control and innovative materials, enable secure anchoring with minimal structural loads. Integration with inflatable composite masts and communication hardware demands multidisciplinary solutions—mechanical, electromagnetic, and procedural. ISRU and additive manufacturing further enhance sustainability and operational flexibility. While simulation and conceptual studies are extensive, experimental validation and integration-focused research remain critical for reliable lunar deployment [1] [4] [7] [14].

References

1. Penetration and resistance characteristics of lunar regolith simulant drilling using a coupled MPM–CDEM approach Li, J., Wang, L., Feng, C., (...), Zhang, Y. Journal of Rock Mechanics and Geotechnical Engineering, 2025 https://www.scopus.com/pages/publications/105014284975?origin=scopusAI

2. Stress and displacement propagation in a drilling tube due to rebounding and non-rebounding hammer impact mechanisms Malla, R.B., Shrestha, B., Zacny, K. Earth and Space 2012 - Proceedings of the 13th ASCE Aerospace Division Conference and the 5th NASA/ASCE Workshop on Granular Materials in Space Exploration, 2012 https://www.scopus.com/pages/publications/84874276140?origin=scopusAI

3. Modeling and experimental investigation of drilling into lunar soils Chen, T., Zhao, Z., Wang, Q., Wang, Q. Applied Mathematics and Mechanics (English Edition), 2019 https://www.scopus.com/pages/publications/85058864718?origin=scopusAI

4. Percussive digging systems for planetary research Craft, J., Wilson, J., Chu, P., (...), Davis, K. IEEE Aerospace and Electronic Systems Magazine, 2010 https://www.scopus.com/pages/publications/78649244016?origin=scopusAI

5. DEM simulation of soil-tool interaction under extraterrestrial environmental effects Jiang, M., Xi, B., Arroyo, M., Rodriguez-Dono, A. Journal of Terramechanics, 2017 https://www.scopus.com/pages/publications/85011392383?origin=scopusAI

6. Research of thermal characteristics in core sampler of lunar soil with vacuum environment Zhao, Z., Lai, X., Quan, Q., (...), Mo, G. Proceedings of the International Astronautical Congress, IAC, 2013 https://www.scopus.com/pages/publications/84904596829?origin=scopusAI

7. Performance evaluation and selection of the core drill bits for lunar rock Wang, X., Wu, Y., Gao, Z., (...), Gao, M. Geomechanics and Geophysics for Geo-Energy and Geo-Resources, 2025 https://www.scopus.com/pages/publications/105007226872?origin=scopusAI

8. Improving tribological performance of polycrystalline diamond compacts by bionic laser texturing and self-lubrication for lunar extreme environments Sun, X., Chen, L., Jia, R., (...), Liu, B. Tribology International, 2027 https://www.scopus.com/pages/publications/105043446155?origin=scopusAI

9. Circular additive manufacturing of recycled PEKK–regolith composites for sacrificial structures in lunar in-situ resource utilization Malekpour, F., Hojjati, M. Composites Part B: Engineering, 2026 https://www.scopus.com/pages/publications/105045842847?origin=scopusAI

10. Multiloop balanced bridge feedback in application to precision pointing Lurie, B.J. International Journal of Control, 1990 https://www.scopus.com/pages/publications/0025418426?origin=scopusAI

11. ACTIVE DAMPING CONTROL DESIGN FOR THE MAST FLIGHT SYSTEM. Ham, Fredric M., Greeley, Scott W. Proceedings of the American Control Conference, 1987 https://www.scopus.com/pages/publications/0023562241?origin=scopusAI

12. Conductive Coatings for EMI/RFI Shielding Ling, J.H. 8th International Wroclaw Symposium on Electromagnetic Compatibility, EMC 1986, 1986 https://www.scopus.com/pages/publications/105003627584?origin=scopusAI

13. FENCE: Flexible Electric Noise Reduction Endo-Shield for the Suppression of Electromagnetic Interference in Low-Field MRI Pfitzer, J., Uecker, M., Scharfetter, H. NMR in Biomedicine, 2026 https://www.scopus.com/pages/publications/105036887872?origin=scopusAI

14. DESIGN OF INFLATABLE MULTI-PURPOSE TOWER FOR SUPPORT OF ROBOTIC AND CREWED LUNAR SURFACE OPERATIONS Testi, C., Bailey, K., Shah, K., (...), Bannova, O. Proceedings of the International Astronautical Congress, IAC, 2024 https://www.scopus.com/pages/publications/85219613108?origin=scopusAI

15. Research on Low-Force Cutting Load Characteristics for Lunar Rock Simulant Wang, Y., Li, P., Xu, B., Liu, J. Proceedings of SPIE - The International Society for Optical Engineering, 2022 https://www.scopus.com/pages/publications/85128270647?origin=scopusAI

16. Biomimetic transfer of plant roots for planetary anchoring Seidl, T., Mugnai, S., Corradi, P., (...), Mancuso, S. International Astronautical Federation - 59th International Astronautical Congress 2008, IAC 2008, 2008 https://www.scopus.com/pages/publications/77950524434?origin=scopusAI

17. Rotary-percussive deep drill for planetary applications Paulsen, G., Zacny, K., McKay, C., (...), Maksymuk, M. Proceedings of the 12th International Conference on Engineering, Science, Construction, and Operations in Challenging Environments - Earth and Space 2010, 2010 https://www.scopus.com/pages/publications/78649275764?origin=scopusAI

18. Helical anchors: Lateral resistance for shaping the lunar surface Crabtree, J., Klosky, J.L., Gash, R., Miller, D. Earth and Space Conference 2008: Proceedings of the 11th Aerospace Division International Conference on Engineering, Science, Construction, and Operations in Challenging Environments, 2008 https://www.scopus.com/pages/publications/70349383851?origin=scopusAI

19. Energy transfer in percussive rock destruction-I. Comparison of percussive methods Lundberg, B. International Journal of Rock Mechanics and Mining Sciences and, 1973 https://www.scopus.com/pages/publications/49549164485?origin=scopusAI

20. LunarVader: Testing of a 1 meter lunar drill in a 3.5 meter vacuum chamber and in the Antarctic lunar analog site Zacny, K., Craft, J., Davila, A., (...), Pollard, W. IEEE Aerospace Conference Proceedings, 2012 https://www.scopus.com/pages/publications/84861131943?origin=scopusAI

21. Energy transmission of down-hole hammer tool and its conditionality Li, X.-B., Rupert, G., Summers, D.A. Transactions of Nonferrous Metals Society of China (English Edition), 2000 https://www.scopus.com/pages/publications/33747660341?origin=scopusAI

22. A piezoelectric-driven ultrasonic/sonic driller for planetary exploration Quan, Q., Li, H., Jiang, S., (...), Tang, D. 2013 IEEE International Conference on Robotics and Biomimetics, ROBIO 2013, 2013 https://www.scopus.com/pages/publications/84898801930?origin=scopusAI

23. Development of a rotary-percussive ultrasonic drill for extraterrestrial rock sampling Quan, Q., Bai, D., Zhao, Z., (...), Wang, Y. IEEE International Ultrasonics Symposium, IUS, 2017 https://www.scopus.com/pages/publications/85039413864?origin=scopusAI

24. Understanding How Speed, Tractive Effort, Digging Height, and Rake Angle Affect Bucket Penetration and Resistive Forces for Rubber Tire Loaders Ur Rehman, A., Awuah-Offei, K. Mining, Metallurgy and Exploration, 2020 https://www.scopus.com/pages/publications/85089782208?origin=scopusAI

25. Discrete element modeling of scaled bucket excavation Ur Atta, R., Awuah-Offei, K., Sherizadeh, T. 54th U.S. Rock Mechanics/Geomechanics Symposium, 2020 https://www.scopus.com/pages/publications/85097953545?origin=scopusAI

26. Experiment study and numerical simulation on the dynamic rock-breaking process of rotary percussive drilling with different cyclic and stress-wave frequencies Xi, Y., Yao, Y., Zha, C., (...), Tian, Y. Petroleum Science and Technology, 2024 https://www.scopus.com/pages/publications/85158882144?origin=scopusAI

27. Study on Rock-Breaking Mechanism of PDC Bits in Rotary Percussive Drilling Based on HJC Model Wang, H., Ji, Z., Su, Q. Shiyou Zuantan Jishu / Petroleum Drilling Techniques, 2025 https://www.scopus.com/pages/publications/105023644471?origin=scopusAI

28. Review of Discrete Element Method Simulations of Soil Tillage and Furrow Opening Aikins, K.A., Ucgul, M., Barr, J.B., (...), Desbiolles, J.M.A. Agriculture (Switzerland), 2023 https://www.scopus.com/pages/publications/85151945925?origin=scopusAI

29. Experimental investigation and theoretical modeling of cutting mechanics using serrated tines in drilling lunar regolith simulant Zhang, Y., Wu, R., Long, Z., (...), Zhang, T. Advances in Space Research, 2024 https://www.scopus.com/pages/publications/85198201680?origin=scopusAI

30. Extremum-seeking control for an Ultrasonic/Sonic Driller/Corer (USDC) driven at high-power Aldrich, J., Sherrit, S., Bao, X., (...), Chang, Z. Proceedings of SPIE - The International Society for Optical Engineering, 2006 https://www.scopus.com/pages/publications/33745816028?origin=scopusAI

31. Integrated Rotor Compensation and Adaptive Stiffness Impedance Control Method for UMD Systems in Soil Sampling Tasks Rao, J., An, J., She, J., Terano, T. IEEE Transactions on Control Systems Technology, 2026 https://www.scopus.com/pages/publications/105020959759?origin=scopusAI

32. Real-time prediction of drilling forces inside lunar regolith based on recurrent neural networks Xu, J., Yuan, X., Zhang, Y., (...), Ding, X. Acta Astronautica, 2022 https://www.scopus.com/pages/publications/85138451024?origin=scopusAI

33. Planar cone penetration resistance in granular matter under low gravity Yang, S., Cheng, X., Hou, M., (...), Chen, L. Physics of Fluids, 2026 https://www.scopus.com/pages/publications/105046897248?origin=scopusAI

34. STUDY ON THE INFLUENCE OF ROTATION SPEED ON DRILLING PERFORMANCE OF LUNAR SOIL SAMPLING DRILLS CAO, J.-F., LIU, X.-L., LIU, W.-J., (...), ZHANG, Y.-B. Thermal Science, 2025 https://www.scopus.com/pages/publications/105008173484?origin=scopusAI

35. Experimental proof-of-concept of the effect of inlet geometry on excavation forces and their reduction for small-scale continuous excavators Just, G.H., Roy, M.J., Joy, K.H., Smith, K.L. Acta Geotechnica, 2024 https://www.scopus.com/pages/publications/85162041577?origin=scopusAI

36. DEM modelling of cone penetration tests in lunar soil Jiang, M., Zhao, T., Wang, X. Granular Matter, 2022 https://www.scopus.com/pages/publications/85119347480?origin=scopusAI

37. Distinct element simulation of lugged wheel performance under extraterrestrial environmental effects Jiang, M., Liu, F., Shen, Z., Zheng, M. Acta Astronautica, 2014 https://www.scopus.com/pages/publications/84897756733?origin=scopusAI

38. Effects of adhesion and particle shape on mechanical behaviors of lunar regolith under low stress condition-3D DEM study Luo, A., Cui, Y., Nie, J., Wang, G. Computers and Geotechnics, 2024 https://www.scopus.com/pages/publications/85201466775?origin=scopusAI

39. Mesoscale Mechanisms Governing the Shear Strength of Lunar Regolith: Effects of Low Confining Stress and Irregular Particle Morphology Chen, J., Li, R., Ji, Y., Mo, P. Materials, 2026 https://www.scopus.com/pages/publications/105035522423?origin=scopusAI

40. Experimental study on the mechanical properties of CAS - 1 Lunar regolith simulant under low stress levels Yu, F., Chen, S., Zhang, Y. Innovative Materials and Design for Sustainable Transportation Infrastructure - Selected Papers from the International Symposium on Systematic Approaches to Environmental Sustainability in Transportation, 2015 https://www.scopus.com/pages/publications/84945373936?origin=scopusAI

41. DEM thermal simulation of bit and object in drilling of lunar soil simulant Cui, J., Hou, X., Wen, G., Liang, Z. Advances in Space Research, 2018 https://www.scopus.com/pages/publications/85048873658?origin=scopusAI

42. VIBRATION CONTROL OF LUNAR ROVER DRILLING SYSTEM IN HARSH ENVIRONMENTS Jang, H.-C., Jeong, H.-R., Kim, J.-H., (...), Lee, Y.-K. Proceedings of the International Congress on Sound and Vibration, 2025 https://www.scopus.com/pages/publications/105044605098?origin=scopusAI

43. Effect of reduced gravity on tool forces of an ultrasonically vibrating blade in regolith simulant Rezich, E., Butt, G., Aubin-Fournier, P.-L., (...), Nguyen, Q. Acta Astronautica, 2025 https://www.scopus.com/pages/publications/105013391888?origin=scopusAI

44. Vibratory excavation and anchoring tools for the lunar surface Klosky, J.L., Sture, S., Ko, H.-Y., Barnes, F. Engineering, Construction, and Operations in Space V, 1996 https://www.scopus.com/pages/publications/84904801515?origin=scopusAI

45. Comparison of the efficiencies of a new class of lunar excavation tools Nathan, Mark, Barnes, Frank, Sture, Stein, (...), Lund, Walter American Society of Mechanical Engineers, Aerospace Division (Publication) AD, 1992 https://www.scopus.com/pages/publications/0026960757?origin=scopusAI

46. A novel lunar soil coring approach based on particle unidirectional flow effect to reduce drag and increase efficiency Chen, H., Li, L., Cui, Y., Jiang, S. Advances in Space Research, 2021 https://www.scopus.com/pages/publications/85103731257?origin=scopusAI

47. Study on the amplitude effect on micro-hole drilling of AISI 4340 by ultrasonic vibration Chen, G., Xu, J., Lian, Z., (...), Yu, H. International Journal of Advanced Manufacturing Technology, 2021 https://www.scopus.com/pages/publications/85109827144?origin=scopusAI

48. Mechanical Properties of Lunar Concrete: Experimental and Simulation Perspectives Pilehvar, S., Kassem, A., Osman, E.H., (...), Kioumarsi, M. Lecture Notes in Civil Engineering, 2026 https://www.scopus.com/pages/publications/105045449182?origin=scopusAI

49. Enhanced in-situ utilization of lunar simulant for fibre-reinforced high-performance concrete: Mechanical properties and cost-effectiveness for lunar applications Shao, R., Chengqing Wu, Li, J. Journal of Materials Research and Technology, 2025 https://www.scopus.com/pages/publications/86000366106?origin=scopusAI

50. Behaviour of regolith-resin-composites subjected to lunar analogue environments Hossain, M.A., Smith, S.T., Tafsirojjaman, T. Journal of Building Engineering, 2026 https://www.scopus.com/pages/publications/105042672800?origin=scopusAI

51. Evaluation of the resistance of composite materials to lunar dust abrasion Boucher, M.L., Potvin, M.-J. 61st International Astronautical Congress 2010, IAC 2010, 2010 https://www.scopus.com/pages/publications/79959457663?origin=scopusAI

52. Fabrication and characterization of DLC coated microdimples on hip prosthesis heads Choudhury, D., Ay Ching, H., Mamat, A.B., (...), Krupka, I. Journal of Biomedical Materials Research - Part B Applied Biomaterials, 2015 https://www.scopus.com/pages/publications/84931954997?origin=scopusAI

53. Biomimetic laser texturing enhances DLC coating adhesion under thermo-mechanical loading Song, B., Li, X., Chen, C., (...), Zhang, Y. Surface and Coatings Technology, 2026 https://www.scopus.com/pages/publications/105052015708?origin=scopusAI

54. Tribological properties of WS₂ coatings deposited on textured surfaces by electrohydrodynamic atomization Li, X., Deng, J., Liu, L., (...), Duan, R. Surface and Coatings Technology, 2018 https://www.scopus.com/pages/publications/85051019439?origin=scopusAI

55. Abrasive wear of PEEK and ATSP-based polymer composite coatings under lunar regolith conditions Bashandeh, K., Lan, P., Polycarpou, A.A. Wear, 2022 https://www.scopus.com/pages/publications/85135792400?origin=scopusAI

56. Towards sustainable lunar habitats with ISRU in Chang'E mission: Mechanical–energy evolution and damage mechanisms of LPBF-printed lunar regolith simulate Li, S., Li, X., Gao, Y., (...), Ding, L. International

Journal of Mining Science and Technology, 2026 https://www.scopus.com/pages/publications/105025401165?origin=scopusAI

57. In-situ Manufacturing and Utilization of Lunar Regolith for Fabrication/ Construction on the Lunar Surface: A Review Zhang, R., Wang, T., Li, L., Wang, G. Yuhang Xuebao/Journal of Astronautics, 2024 https://www.scopus.com/pages/publications/85201626164?origin=scopusAI

58. Regolith-based additive manufacturing for sustainable development of lunar infrastructure – An overview Isachenkov, M., Chugunov, S., Akhatov, I., Shishkovsky, I. Acta Astronautica, 2021 https://www.scopus.com/pages/publications/85099481284?origin=scopusAI

59. Characterization and optimization of vibration-assisted drilling of fibre reinforced epoxy laminates Sadek, A., Attia, M.H., Meshreki, M., Shi, B. CIRP Annals - Manufacturing Technology, 2013 https://www.scopus.com/pages/publications/84878855787?origin=scopusAI

60. Delamination in surface machining of CFRP composites using rotary ultrasonic machining with horizontal ultrasonic vibration Wang, H., Zhang, D., Li, Y., (...), Burks, A.R. ASME 2020 15th International Manufacturing Science and Engineering Conference, MSEC 2020, 2020 https://www.scopus.com/pages/publications/85101474807?origin=scopusAI

61. Mechanisms of machining and damage suppression in thermoset carbon fiber composites via robotic ultrasonic-assisted drilling Wang, C.-C., Chien, Y.-H., Shen, M.-Y., Liu, C.-H. International Journal of Advanced Manufacturing Technology, 2026 https://www.scopus.com/pages/publications/105046391455?origin=scopusAI

62. Rotary ultrasonic machining of woven CFRP composite in a cryogenic environment Thirumalai Kumaran, S., Ko, T.J., Li, C., (...), Uthayakumar, M. Journal of Alloys and Compounds, 2017 https://www.scopus.com/pages/publications/85007300190?origin=scopusAI

63. System concept for in-situ casing generation in a deep Lunar borehole Linossier, A.R. Proceedings of the International Astronautical Congress, IAC, 2017 https://www.scopus.com/pages/publications/85051361664?origin=scopusAI

64. Modeling and Feedback Control for a Guyed, Flexible, Tubular Lunar Tower Portmann, V.P., Miller, A.S., Zhang, J.Z., (...), Chin, H.H. IEEE Aerospace Conference Proceedings, 2024 https://www.scopus.com/pages/publications/85193785963?origin=scopusAI

65. Integrating topside design Scott, R. Jane's Navy International, 2005 https://www.scopus.com/pages/publications/23944431511?origin=scopusAI

66. Software tools and approaches to enable international lunar surface operations O'Neil, P.G. 60th International Astronautical Congress 2009, IAC 2009, 2009 https://www.scopus.com/pages/publications/77953631277?origin=scopusAI

67. Systems Integration Support for NASA’s Lunar Surface Innovation Initiative Clyde, B., Hurley, D., Bussey, B. Accelerating Space Commerce, Exploration, and New Discovery conference, ASCEND 2022, 2022 https://www.scopus.com/pages/publications/105005570828?origin=scopusAI

68. Borehole reinforcement based on polymer materials induced by liquid-gas phase transition in simulating lunar coring Mo, D., Liu, T., Zhao, Z., (...), Xie, H. International Journal of Mining Science and Technology, 2025 https://www.scopus.com/pages/publications/105001498628?origin=scopusAI

69. Ultra-lightweight cement slurry to seal wellbore of poor wellbore stability Kremieniewski, M. Energies, 2020 https://www.scopus.com/pages/publications/85087969762?origin=scopusAI

70. Design approaches to enhance electromagnetic compatibility of aerospace ground support systems Ockene, N.J. IEEE Transactions on Aerospace and Electronic Systems, 1966 https://www.scopus.com/pages/publications/84939065501?origin=scopusAI

71. Diversity Techniques for Multipath Fading Channels between Earth and the Moon Divsalar, D., Net, M.S., Cheung, K.-M., (...), Tamrat, A. IEEE Aerospace Conference Proceedings, 2023 https://www.scopus.com/pages/publications/85160535262?origin=scopusAI

72. Performance Assessment of Magnetic Polymeric Composites for EMI Reduction in Small Satellites Casling, A.N., Marra, F., Tamburrano, A., (...), Scione, E. Proceedings of 2025 ESA Workshop on Aerospace EMC, Aerospace EMC 2025, 2025 https://www.scopus.com/pages/publications/105012249882?origin=scopusAI

73. Feedback control Hong, K.-S., Shah, U.H. Advances in Industrial Control, 2019 https://www.scopus.com/pages/publications/85061369392?origin=scopusAI

74. Integrated intelligent control of flexible structures - Analysis and experiments Bilimoria, K.D., Garrett, F.E., Livneh, R., Schmidt, D.K. Proceedings of SPIE - The International Society for Optical Engineering, 1993 https://www.scopus.com/pages/publications/85076060585?origin=scopusAI

75. Robustness comparative study of two control schemes for 3-DOF flexible manipulators Feliu, V., Somolinos, J.A., García, A., Sánchez, L. Journal of Intelligent and Robotic Systems: Theory and Applications, 2002 https://www.scopus.com/pages/publications/0036691823?origin=scopusAI

76. Autonomous in-situ resource utilization of lunar water ice enabled by a permanently operating lunar exploration rover (POLER) Huschke, A., Tonina, T., Mani, V. Proceedings of the International Astronautical Congress, IAC, 2023 https://www.scopus.com/pages/publications/85188010968?origin=scopusAI

77. Mission Concepts for Permanently Shadowed Regions in the Lunar South Pole Papworth, J. Proceedings of the International Astronautical Congress, IAC, 2025 https://www.scopus.com/pages/publications/105032955521?origin=scopusAI

78. The Technical Feasibility of 3D Printing Technology for Lunar Base Settlement Kaur, G., Wisran, B.A.S.A.M., AlGharrash, A.I. Proceedings of the International Astronautical Congress, IAC, 2023 https://www.scopus.com/pages/publications/85187988619?origin=scopusAI

79. Outpost robotic technologies for lunar surface prospecting and processing Glass, B.J., Lee, K., Cannon, H., (...), Huffman, S. International Astronautical Federation - 58th International Astronautical Congress 2007, 2007 https://www.scopus.com/pages/publications/54949119958?origin=scopusAI

80. Innovative Regolith Transport Systems for Extreme Lunar and Space Conditions Salihler, S., Kasanya, B. Proceedings of the International Astronautical Congress, IAC, 2024 https://www.scopus.com/pages/publications/85219587868?origin=scopusAI

81. From importing to exporting: The impact of ISRU on space logistics Bienhoff, D.G. AIAA SPACE Conference and Exposition 2011, 2011 https://www.scopus.com/pages/publications/84880636122?origin=scopusAI